

LAB 5 – Struct and Enum

1. Basic struct - Student

```
struct Student
{
    public int Id;
    public string Name;
    public double Marks;
}
class Program
{
    static void Main(string[] args)
    {
        Student s;
        s.Id = 102;
        s.Name = "Rajesh";
        s.Marks = 90.5;
        Console.WriteLine("Student Details :-");
        Console.WriteLine($"Id : {s.Id}");
        Console.WriteLine($"Name : {s.Name}");
        Console.WriteLine($"Marks : {s.Marks}");
    }
}
```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab5\$ dotnet run

Student Details :-

Id : 102

Name : Rajesh

Marks : 90.5

2. Basic struct – Point

```
struct Point
{
    public int X, Y;
}
class Program
{
    static void Main(string[] args)
    {
        Point p1;
        p1.X = 10;
        p1.Y = 20;

        Point p2;
        p2.X = 30;
        p2.Y = 40;

        Console.WriteLine($"p1 = {p1.X}, {p1.Y}");
        Console.WriteLine($"p2 = {p2.X}, {p2.Y}");
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab5$ dotnet run
p1 = 10, 20
p2 = 30, 40
```

3. Struct using array

```
struct Book
{
    public string Title;
    public double Price;
}
class Program
{
    static void Main(string[] args)
    {
        Book[] books = new Book[3];
        books[0].Title = "C#";
        books[0].Price = 100;
        books[1].Title = "Java";
        books[1].Price = 200;
        books[2].Title = "Python";
        books[2].Price = 300;
        Console.WriteLine("Books Details:-");
        foreach (Book book in books)
        {
            Console.WriteLine($"Title:{book.Title}, Price:{book.Price}");
        }
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab5$ dotnet run
Books Details:-
Title:C#, Price:100
Title:Java, Price:200
Title:Python, Price:300
```

4. Enum – Day

```
enum Day
{
    Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday
}
class Program
{
    static void Main(string[] args)
    {
        Day day = Day.Tuesday;
        Console.WriteLine($"Today is {day}");
        Console.WriteLine($"Underlying value is {(int)day}");
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab5$ dotnet run
Today is Tuesday
Underlying value is 2
```

5. Enum – OrderStatus and Struct – Order

```
enum OrderStatus
{
    Pending,
    Processing,
    Shipped,
    Delivered
}

struct Order
{
    public int OrderId;
    public OrderStatus Status;
}

class Program
{
    static void Main(string[] args)
    {
        Order[] orders = new Order[3];
        orders[0].OrderId = 1011;
        orders[0].Status = OrderStatus.Pending;
        orders[1].OrderId = 1012;
        orders[1].Status = OrderStatus.Processing;
        orders[2].OrderId = 1013;
        orders[2].Status = OrderStatus.Shipped;
        Console.WriteLine("Order details:-");
        foreach (var order in orders)
        {
            Console.WriteLine($"Order Id: {order.OrderId} Status: {order.Status}");
        }
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab5$ dotnet run
Order details:-
Order Id: 1011 Status: Pending
Order Id: 1012 Status: Processing
Order Id: 1013 Status: Shipped
```

6. Enum – Status and Struct – Task

```
enum Status
{
    Pending,
    InProgress,
    Completed
}

struct Task
{
    public string Title;
    public Status Status;
}

class Program
{

```

```

static void Main(string[] args)
{
    Task t1;
    t1.Title = "Task 1";
    t1.Status = Status.Pending;

    Task t2;
    t2.Title = "Task 2";
    t2.Status = Status.InProgress;

    Task t3;
    t3.Title = "Task 3";
    t3.Status = Status.Completed;

    Console.WriteLine("Task list:-");
    PrintTask(t1);
    PrintTask(t2);
    PrintTask(t3);
}
static void PrintTask(Task t)
{
    Console.WriteLine($"Task: {t.Title}");
    Console.WriteLine($"Status: {t.Status}");
    Console.WriteLine("-----");
}
}

```

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Task list:-

Task: Task 1

Status: Pending

Task: Task 2

Status: InProgress

Task: Task 3

Status: Completed
